

The empirical recurrence for OEIS A236114: recovering a conjecture that is not written down

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Abstract

OEIS A236114 records “Empirical recurrence of order 84”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 2248$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 84, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 84 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A236114 is “Number of $(n+1) \times (2+1)$ 0..3 arrays colored with the difference of the maximum and the upper median in each 2×2 subblock.”. It has offset 1 and begins

2248, 35468, 542140, 8253392, 126117020, 1934727612, 29594969044, 452447788292, ...

The entry states:

Empirical recurrence of order 84 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 09:06:21, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 2248$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 2248$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 4496$ exact terms of the model returns order 84 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{84} c_i a(n-i),$$

c_1	0	c_{22}	75336887927798	c_{43}	35618938905028099250	c_{64}	365032193946900
c_2	12	c_{23}	−3188160913316032	c_{44}	35270764427477899281	c_{65}	−28448146160355
c_3	848	c_{24}	−4814737687231849	c_{45}	−152463896537375976630	c_{66}	−6279421863344
c_4	25381	c_{25}	11126835057007680	c_{46}	−44664273774898704956	c_{67}	−28833997393341
c_5	142584	c_{26}	11129663333854529	c_{47}	114813030118314182394	c_{68}	15726295207955
c_6	891665	c_{27}	31508945648227214	c_{48}	−62502980305846709977	c_{69}	10573503840829
c_7	4586174	c_{28}	71538578588025657	c_{49}	395058212223326297850	c_{70}	43957343603851
c_8	−32149834	c_{29}	−91218877747075094	c_{50}	514151115742934646089	c_{71}	58231802925235
c_9	−177185788	c_{30}	−223510785416188891	c_{51}	−824598365214115488552	c_{72}	31542383674334
c_{10}	−63128417	c_{31}	−262061430211982394	c_{52}	−551707832589473269464	c_{73}	16327195531060
c_{11}	−2229633856	c_{32}	−520048101337101499	c_{53}	352170961613093842932	c_{74}	−2929548818667
c_{12}	12797946554	c_{33}	274439336951232950	c_{54}	−675959843661013725119	c_{75}	−4307094629305
c_{13}	117113142660	c_{34}	2013934527635032563	c_{55}	5690004897205093356	c_{76}	−5725342386078
c_{14}	−137897212510	c_{35}	481715071074032748	c_{56}	988307137864272302263	c_{77}	10831013202201
c_{15}	−573109363522	c_{36}	−1127392624063272089	c_{57}	−534876454977120528834	c_{78}	4654501772317
c_{16}	701751184162	c_{37}	3950219736432769236	c_{58}	−37857824879335836011	c_{79}	3629682257697
c_{17}	−15169835815996	c_{38}	−452241245171671934	c_{59}	928515880713666092574	c_{80}	−365867690632
c_{18}	−10652045429654	c_{39}	−6230783313159745030	c_{60}	−1219344135073258721768	c_{81}	−178991341568
c_{19}	185442147194072	c_{40}	15593320667201101072	c_{61}	−1360599278032114534622	c_{82}	−13468364845
c_{20}	101448123714621	c_{41}	−3683160515944184396	c_{62}	1093561329022347852548	c_{83}	230537427402
c_{21}	−398721344820884	c_{42}	−45476943995021351180	c_{63}	1424754237912443367440	c_{84}	−4365887815

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 84, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $84 + 4$ terms removed returns order 84 as well, so $\deg q_{\min} = 84 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $84 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 84 that this sequence satisfies — and that family turns out to have exactly one member.

References

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